1. **Title Page / Cover Page:**

University Clinic Appointment System

Principles of Software Engineering:

Module Code: ITS 610204 / ITS 64604

Sustainable Development Goal:

SDG 3 – Good Health and Well-Being

Team Members:

ZHAO ZISHUO 0384054 – Lead Developer (GitHub C Prototype)

Georgius Bertrand Raphael Wencelas 0387424 – Scrum Master (Trello C Agile Process & PowerPoint)

LU YINGWEN 0388085 – Systems Analyst (UML C Testing) XIANG ZIYANG 0384280 – UI/UX Designer (Design C Mockups) Khaled Barakeh 0382162 – Documentation C QA (Final Report)

Submission Date: 13 March 2026

2. Abstract

The project is a case of a University Clinic Appointment System that is a digital application that will enhance access to healthcare by students in a university setting. The system enables the students to log in, choose a clinical department and make appointments with ease using a user-friendly interface.

The prototype will help to minimize waiting time and enhance the efficiency of the process of appointment scheduling at university clinics. The system was created in line with principles of Agile software development, and project activities were planned with the help of Trello and project documentation was registered in GitHub.

The prototype of the user interface was created with the help of Figma, which displays the key characteristics of the system such as student authentication, selecting a department, booking an appointment and booking confirmation. This project falls under Sustainable Development Goal (SDG) 3 - Good Health and Well-Being that is dedicated towards the development of health services and encouragement of access to healthcare.

3. Project Introduction

University clinics are the most important facilities that offer healthcare services to students, yet most of them use manual systems of booking. Such systems may lead to excessive waiting time, time schedule and ineffective administration of appointments. In response to those difficulties, the current project suggests a University Clinic Appointment System, a web-based platform that will enable students to make appointments through the Internet in a simple way.

With this system, students are able to pick a clinic department in a short time, an available date and time and confirm their appointment. The given solution will help optimize the overall efficiency of healthcare services in the university, where students will have an opportunity to receive medical help more conveniently. The project is in line with Sustainable Development Goal (SDG) 3 - Good Health and Well-Being, which focuses on the enhancement of access to healthcare services and on the use of technology to strengthen the healthcare system of the population.

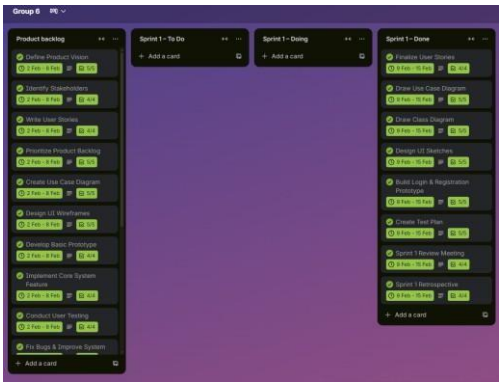
4. **Agile Process / Trello Usage**

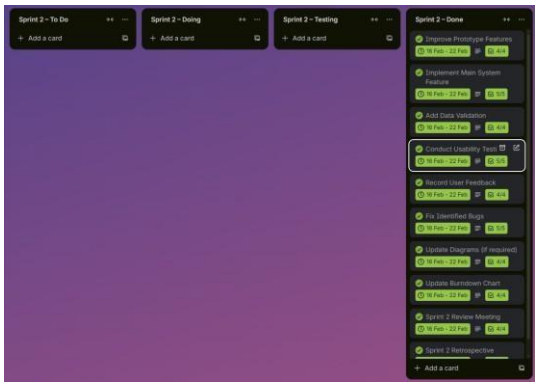
Some of the Agile methodologies that were used in developing the project include Agile focusing on teamwork, flexibility, and continuous improvement during the development process.

Trello was a project management tool used by the team to track and organize the tasks. The Trello board was designed in the form of a Kanban workflow with three columns:

- To Do
- Doing
- Done

All the tasks were defined as cards in Trello and allocated to a team member. Designing the UI screens, constructing the UML diagrams, testing the system functions and recording the project progress were among the tasks. Agile and Trello allowed the team to efficiently managed the tasks, depicted the progress of the project, and easily collaborated in the development process.





Project activities and the advancement of each sprint was planned in the Trello board. Activities that were taken out of the Product Backlog into Do, Doing, and Done as the team finished with the development work.

Project Management Board (Trello):

https://trello.com/b/iHIn6YOr/group-6

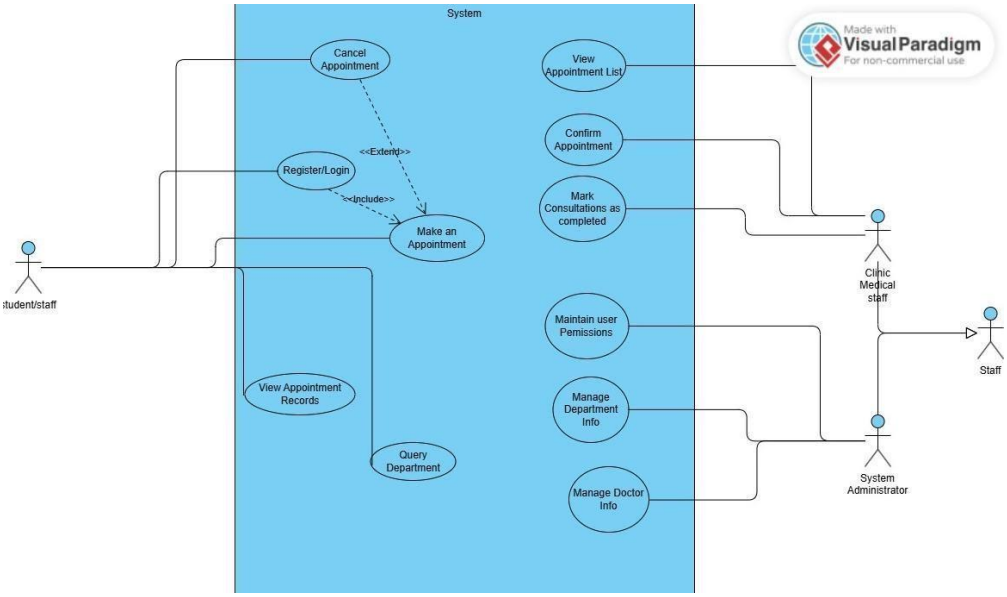
5. SDLC Phases

Planning / Requirements-

In the planning stage, the team has done the identification of the objectives of the project and specified the system requirements. The system will aim at allowing students to have a digital platform where they can make appointments at the university clinic with ease. User story was developed to show how the students will use the system. These user stories assisted the team in knowing the key features needed by the system, including the ability to log in into the system, pick a department, make an appointment and get confirmation. The planning also involved drawing the project backlog and arrangement of works in the Trello board.

Analysis-

During the analysis stage, the system requirements were analyzed further and these were represented based on UML diagrams. The team came up with Use Case Diagram to illustrate the interrelationship between the user and the system. The student is the primary participant in the system as he/she interacts with the system to execute activities like login, choosing a clinic department and make an appointment. The Use Case Diagram assists in drawing the picture of the system functionality and can be applied to make sure that all the needed interactions are well presented.



Class Diagram-

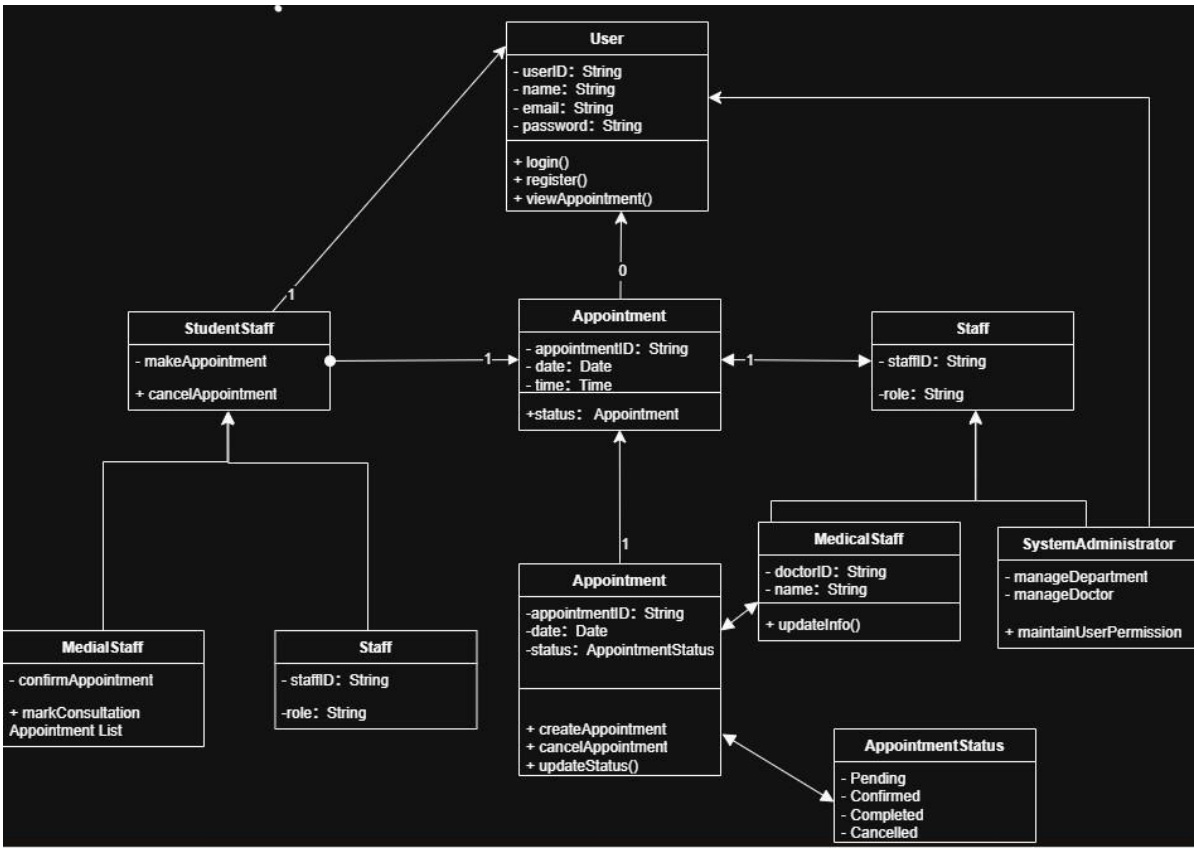
Class Diagram is the structural design of university clinic Appointment System. It depicts the principal types of classes of the system and the connections between them.

The User class is the simplest user object and includes the userID, name, email and password among other simplest operations, which include, logging in, registering and checking appointments. StudentStaff class is a type of students that utilize the system to control their appointments.

Students are able to make appointments and cancel them in case of need. The Appointment class consists of data regarding every clinic appointment such as appointment ID, date, time, and status of the appointment. It also includes operations in creating, canceling and updating the records of the appointments. The Staff type is used to indicate the employees of the clinic that handle appointments and other medical services. This is further divided into MedicalStaff and SystemAdministrator.

The MedicalStaff group is symbol of medical specialists like doctors who have the authority to verify appointments and control consultations. SystemAdministrator class deals with the management of the system by taking care of departments, managing physicians and system user permissions.

Lastly, the AppointmentStatus designates the available appointment statuses such as the Pending, Confirmed, Completed, and Cancelled. All in all, the Class Diagram is used to picture the system structure and the connection between users, appointments, and clinic staff, which are directly involved in the appointment booking process.



Design

At the design stage, the UI/UX designer designed wireframes and interface mockups of the system. The architecture was based on the design of a simple and easy to use interface such that the students can use the system with ease.

The main interface screens include:

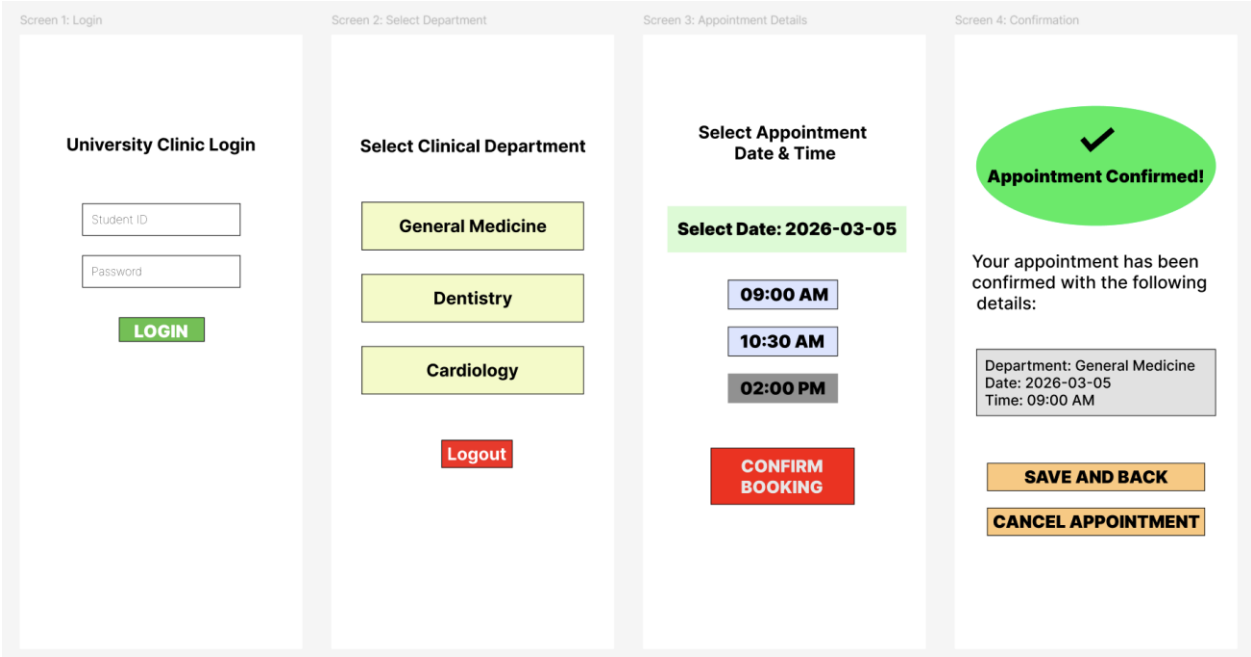
Login screen

Department selection screen

Appointment booking screen

Appointment confirmation screen

These screens were designed to ensure a smooth and intuitive user experience.



Prototype Development

The system prototype was created with the help of Figma, a collaborative design tool that was used to create interactive user interface prototype.

The prototype provides an overview of how the system functions and enables the users to simulate how to book a clinic appointment.

Core Features Implemented

Student authentication (login system)

Clinical department selection

Real-time appointment booking (date and time)

Booking confirmation feedback

These characteristics are the fundamental functionality of a university clinic appointment booking system.

Testing / Feedback

To test functionality and the usability of the system prototype, testing was done. To make sure that the system is acting as it is designed to, the team developed a Test Plan and some Functional Test Cases.

The areas that the testing process was testing included major system functions like:

Login authentication
Department selection
Appointment booking process
Confirmation message display

Testing process revealed that the interface of the system is user-friendly and the key features work properly.

Test ID	Feature	Action	System Response	Screenshot	Status
TC-01	**User Authentication**	Click "LOGIN"	System navigates to "Select Clinical Department" (Screen 2).		Pass
TC-02	**Dept Selection**	Click "General Medicine"	System navigates to "Appointment Details" (Screen 3).		Pass
TC-03	**Availability Logic**	Observe the "02:00 PM" slot	Slot is greyed out/disabled, reflecting "Slot Full" logic from Sequence Diagram.		Pass
TC-04	**Booking Process**	Click "CONFIRM BOOKING"	System displays "Appointment Successful!" (Screen 4) with correct summary.		Pass
TC-05	**Navigation Loop**	Click "BACK TO DASHBOARD"	System returns to the Select screen.		Pass
TC-06	**Logout**	Click "Logout"	System returns to the Login screen.		Pass

User Feedback

Reviewing of the prototype was conducted as a user so that the interface is not complicated and difficult to use. The process of the login procedure, selection of department, and booking of appointment were made to reduce the effort of users and make the system easy to understand by the students.

Improvement

According to the assessment, there are a number of changes that can be discussed in the further development which include the addition of the appointment reminder messages, the further development of slot availability management, and the better usability of the interface.

6. GitHub Evidence

Project Repository (GitHub): https://github.com/15128595106zzs-hub

GitHub was used as the main platform for managing the project repository and documenting the project development process.

The repository was organized into several folders corresponding to different phases of the Software Development Life Cycle:

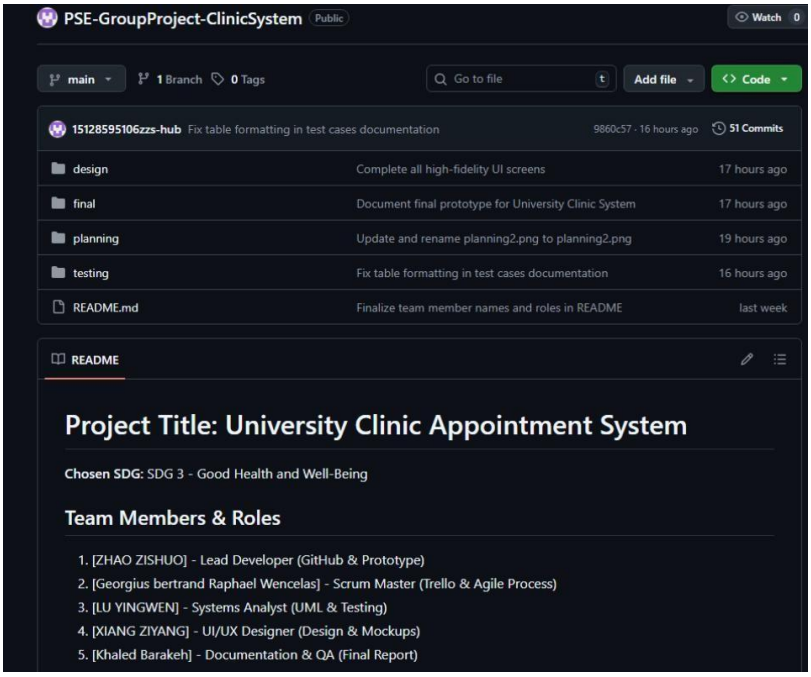
Planning – project planning documents

Design – UI designs and UML diagrams

Testing – test plans and test cases

Final – final prototype documentation and report

Frequent commitments took place during the whole project process to keep track of the updates and to prove the progress of the team during development.



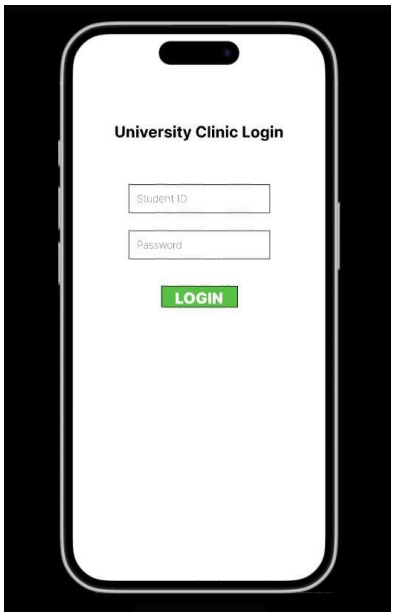
7. **Prototype Showcase**

The prototype shows the main characteristics of the University Clinic Appointment System.

Interactive Prototype Link:
https://www.figma.com/proto/wY9MFhTtDePozUnzokEvXx/Untitled?node-id=1-4&p=f&t=8yzcByiFRai5wcwa-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A4

Login Interface

The system has a login screen through which students can use their student ID and password to gain entry to the system.

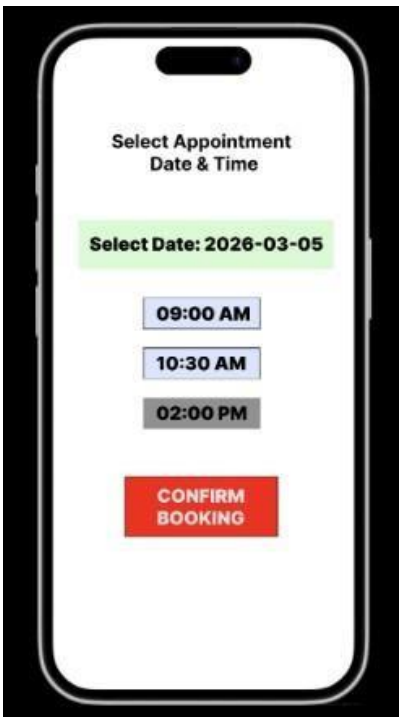


Department Selection

Once the students enter, they are given a choice of the clinics department of their choice.

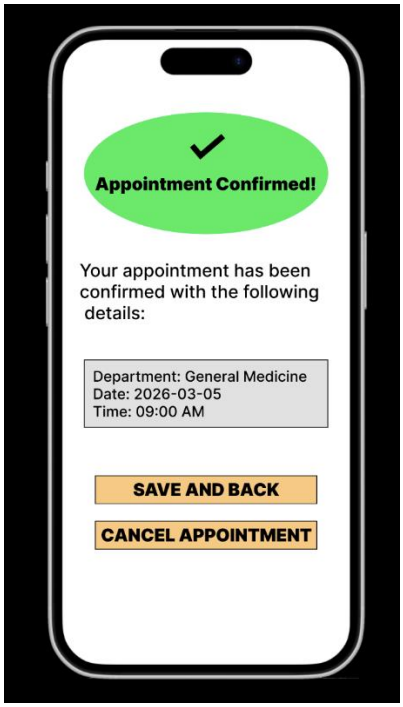
Appointment Booking

The student is able to select a date and time of appointment.



Booking Confirmation

The system will provide the user with a confirmation message after the appointment is booked.



8. Reflections s Lessons Learned

The project provided the team with invaluable experience in the use of concepts of software engineering to a practical issue.

This project made the team realize the need of :

Agile development practices

Team collaboration

Version control using GitHub

Task management using Trello

User interface design

System testing and evaluation

Collaborative team work also enhanced team members communication and problem solving abilities.

9. Conclusion

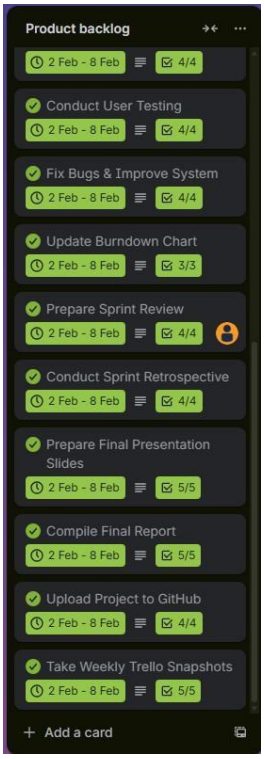
Conclusion The prototype of the University Clinic Appointment System shows that technology can enhance healthcare services in a university setting.

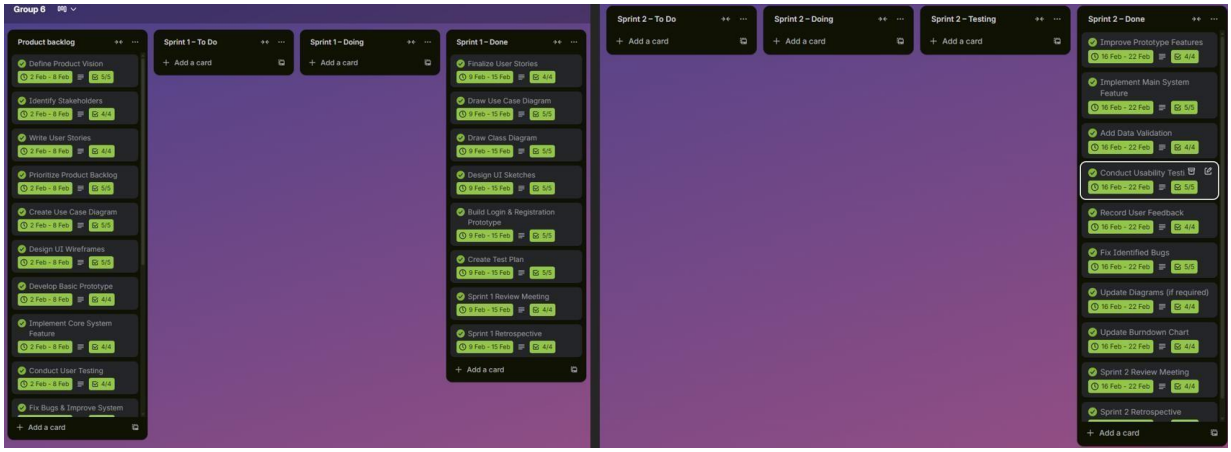
The system enables students to make appointments online and therefore saves time in waiting as well as it enhances efficiency in scheduling the clinic. Agile development practices and software development tools were effectively used in the project to design a working system prototype, test it. As it grows, the system can be developed into a full-fledged application that facilitates legitimate university healthcare services.

Appendices

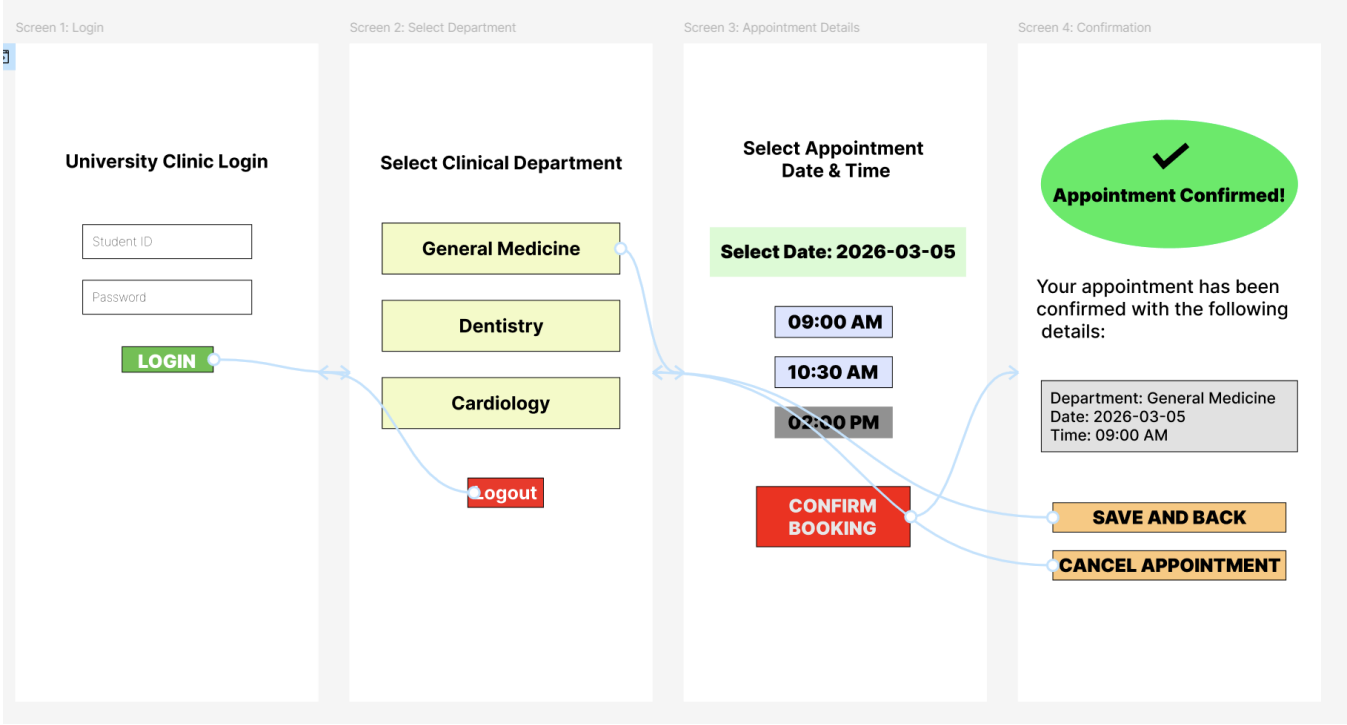
Screenshots from GitHub repository

Trello board screenshots

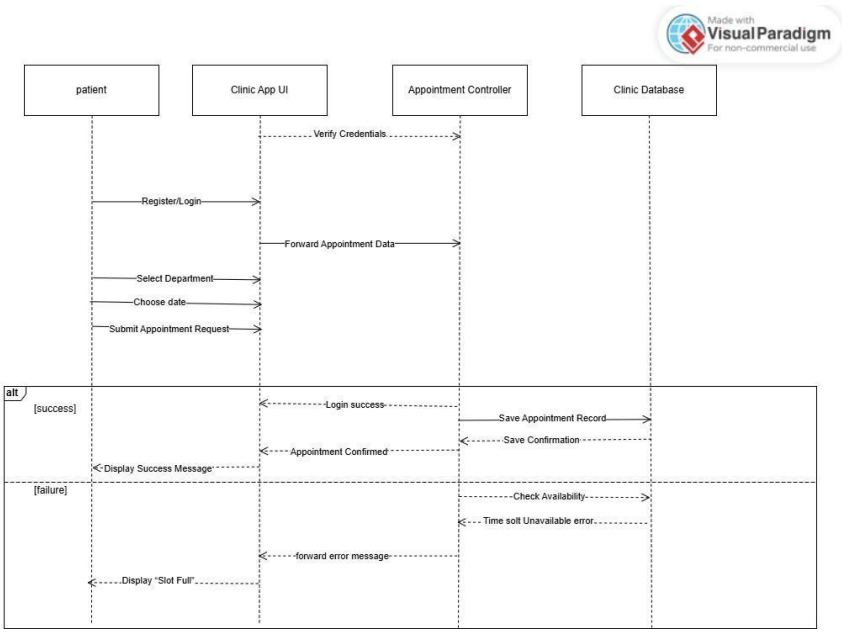




UI prototype screens



Testing evidence

Test ID	Feature	Action	System Response	Screenshot	Status
TC-01	User Authentication	Click "LOGIN"	System navigates to "Select Clinical Department" (Screen 2).		Pass
TC-02	Dept Selection	Click "General Medicine"	System navigates to "Appointment Details" (Screen 3).		Pass
TC-03	Availability Logic	Observe the "02:00 PM" slot	Slot is greyed out/disabled, reflecting "Slot Full" logic from Sequence Diagram.		Pass
TC-04	Booking Process	Click "CONFIRM BOOKING"	System displays "Appointment Successful!" (Screen 4) with correct summary.		Pass
TC-05	Navigation Loop	Click "BACK TO DASHBOARD"	System returns to the Select screen.		Pass
TC-06	Logout	Click "Logout"	System returns to the Login screen.		Pass

UML diagrams